CHAPTER 4

Differentiation, convexity and continuity

S29, S31, S32; revisited in S33 and S40

Objectives

Differentiate a composite function; study variations and convexity; prove existence and uniqueness; approximate a solution.

Course notes

Differentiating a composite function

If u is differentiable on I and v is differentiable on an interval containing $u(I)$, then $(v \circ u)' = (v' \circ u)u'$. The domains must be checked before the calculation.

$$(u^n)' = nu'u^{n-1}, \quad (e^u)' = u'e^u, \quad \left(\frac{1}{u}\right)' = -\frac{u'}{u^2}, \quad (\sqrt{u})' = \frac{u'}{2\sqrt{u}} \quad (u > 0).$$

For example, $f(x) = \sqrt{3x+1}$ is differentiable for $x > -\frac{1}{3}$ and $f'(x) = \frac{3}{2\sqrt{3x+1}}$. The sign of f' determines the variations of f; the sign of f does not.

Convexity

A function f is convex on an interval I if its curve lies below its chords: for $x, y \in I$ and $t \in [0,1]$,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y).$$

For a twice-differentiable function, the following equivalences are accepted: f is convex; f' is increasing; $f'' \geq 0$; the curve lies above all its tangents. The inequalities are reversed for a concave function. A point of inflection is a point where the curve changes convexity.

Proof: position relative to the tangents. Suppose $f'' \geq 0$. Fix a and set $g(x) = f(x) - f(a) - f'(a)(x-a)$. Since f' is increasing, $g'(x) = f'(x) - f'(a) \leq 0$ for $x \leq a$ and ≥ 0 for $x \geq a$. Thus g attains its minimum $g(a) = 0$ at a, so $f(x) \geq f(a) + f'(a)(x-a)$.

Examples. x^4 is convex and $f''(0) = 0$, but 0 is not the abscissa of a point of inflection. x^3 changes convexity at 0. The convexity of the exponential and its tangent at 0 give $e^x \geq 1 + x$.

Continuity and equations

f is continuous at a if $\lim_{x \to a} f(x) = f(a)$. Every differentiable function is continuous. The converse is false: $|x|$ is continuous at 0 but not differentiable at 0.

Intermediate value theorem

If f is continuous on $[a,b]$, every real number between $f(a)$ and $f(b)$ has at least one preimage in $[a,b]$. If f is moreover strictly monotonic, this preimage is unique. One may also work on an open interval using the limits at the endpoints.

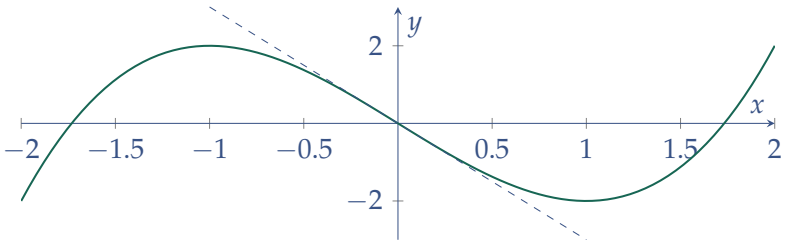
Worked example. $f(x) = x^3 + x - 1$ is continuous on $[0,1]$. Since $f(0) = -1$ and $f(1) = 1$, the IVT gives a zero in $]0,1[$. Since $f'(x) = 3x^2 + 1 > 0$, it is unique. $f(0.68) < 0 < f(0.69)$ provides a bracketing, without replacing the justifications of existence and uniqueness.

A piecewise-defined function. For $x \neq 1$, set $g(x) = \frac{x^2-1}{x-1}$, and set $g(1) = a$. Since $g(x) = x + 1$ when $x \neq 1$, $\lim_{x\to 1} g(x) = 2$. The function is continuous at 1 if and only if $a = 2$. The value at the point must therefore be compared with the limit, and not replaced by it without justification.

Sequences and continuity. If $u_n \to \ell$ and f is continuous at ℓ, then $f(u_n) \to f(\ell)$. For a convergent recursively defined sequence $u_{n+1} = f(u_n)$, we obtain $\ell = f(\ell)$.

Bisection

For f continuous and $f(a)f(b) < 0$, compute $m = \frac{a+b}{2}$. If $f(m) = 0$, stop; otherwise keep the half whose endpoints have images of opposite signs. After n steps, the length is $\frac{b-a}{2^n}$. The midpoint gives an error of at most half the length. The method does not guarantee uniqueness; this is proved separately.



$y = x^3 - 3x$ and its tangent at the point of inflection.

Research activity

Derivatives and reading graphs — 35 min.

Consider the function f defined on $\mathbb{R}$ by $f(x) = x^3 - 3x$. Denote its graph by $\mathcal{C}_f$.

1. Calculate $f'(x)$ and $f''(x)$, then draw up their sign tables.
2. Deduce the variations of f and its intervals of convexity and concavity.
3. Determine the coordinates of the point of inflection of $\mathcal{C}_f$ and an equation of the tangent at this point.
4. Using the limits, the extrema and the intermediate value theorem, determine the number of real solutions of $f(x) = 1$.
5. Explain, from this example, why continuity is not enough to ensure that a solution is unique.

Progressive exercises

E1 · Practice. Define on $\mathbb{R}$ the functions $f(x) = (2x-1)^5$ and $g(x) = e^{-x^2}$. Calculate $f'(x)$ and $g'(x)$.

E2 · Practice. Let f be the function defined on $\mathbb{R}$ by $f(x) = x^4 - 6x^2$. Determine the intervals on which f is convex or concave.

E3 · Practice. Let $\mathcal{C}$ be the graph of the exponential function. Determine an equation of its tangent at the point with abscissa 1. Using convexity, deduce an inequality valid for every real number x.

E4 · Practice. Prove that the equation $x^3 + 2x = 5$ has a unique real solution. Bracket this solution between two consecutive integers.

E5 · Practice. A bisection algorithm starts from the interval $[0;2]$ and halves its length at each step. Determine the minimum number of steps guaranteeing an interval of length at most 10^{-3}.

Textbook or past-paper work

Barbazo, p. 132, exercise 29. In class after the study of convexity: questions 1 and 2. At home: finish questions 3 to 5. The aim is to distinguish the sign of f', which governs the variations, from that of f'', which governs convexity.

Homework

Homework 4 — A maximum and an equation — 60 min.

Consider the function f defined on $[0;+\infty[$ by $f(x) = (x+1)e^{-x}$. Its graph is denoted $\mathcal{C}_f$.

1. Calculate $f(0)$ and the limit of f at $+\infty$. Calculate $f'(x)$, then draw up the table of variations of f.
2. Calculate $f''(x)$. Study the convexity of f and determine the coordinates of the possible point of inflection of $\mathcal{C}_f$.
3. Prove that the equation $f(x) = \frac{1}{2}$ has a unique solution α on $[0;+\infty[$.
4. Write a Python function that uses bisection on $[0;2]$ and returns two real numbers a and b such that $a \leq \alpha \leq b$ and $b - a \leq 10^{-3}$. Justify the choice of the initial interval.
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with abscissa 0. Is the position of the curve relative to this single tangent enough to state that f is concave on its whole domain? Justify.

Teaching notes and solutions

Teacher notes

The scheme of work condenses this chapter into three sessions: genuinely split S29 between differentiation and convexity, then revisit the proof about tangents in S33. The vocabulary continuous/differentiable/convex must be distinguished by examples. Continuity gives existence via the IVT; strict monotonicity gives uniqueness. For bisection, require an initial interval where the sign changes and a continuous function. A change of convexity, and not merely the vanishing of f'', characterises a point of inflection.

Solutions to the activity

1. $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$ and $f''(x) = 6x$. The sign of f' is positive on $]-\infty, -1[$ and $]1, +\infty[$, zero at $-1, 1$, negative between these values. The sign of f'' is negative to the left of 0, zero at 0, positive to the right.

2. The variations and the limits are summarised by

x	$-\infty$		-1		1		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	$-\infty$	$\nearrow$	2	$\searrow$	-2	$\nearrow$	$+\infty$

   The sign of f'' gives concavity on $]-\infty, 0]$ and convexity on $[0, +\infty[$.

3. The convexity changes at 0, and $f(0) = 0$: the point of inflection is $(0, 0)$. Its tangent has equation $y = f(0) + f'(0)x = -3x$.

4. The function is continuous. On $]-\infty, -1]$, it is strictly increasing from $-\infty$ to 2: the level 1 is reached exactly once. On $[-1, 1]$, it strictly decreases from 2 to -2: a second solution. On $[1, +\infty[$, it strictly increases from -2 to $+\infty$: a third. The points $-1, 1$ are not solutions, so no solution is counted twice.

5. Continuity provides the existence of a preimage for intermediate values. It does not provide uniqueness: the function studied is continuous and the level 1 has three preimages.

E1. For $u = 2x - 1$, $(u^5)' = 5u'u^4 = 10(2x-1)^4$. For $v = -x^2$, $(e^v)' = v'e^v = -2xe^{-x^2}$. Both functions are differentiable on $\mathbb{R}$.

E2. $f' = 4x^3 - 12x$ then $f'' = 12(x^2 - 1)$. The sign of f'' is positive outside $[-1, 1]$, negative inside. So f is convex on $]-\infty, -1]$ and $[1, +\infty[$, concave on $[-1, 1]$. The changes of convexity give the points of inflection $(-1, -5)$ and $(1, -5)$.

E3. The tangent is $y = f(1) + f'(1)(x-1) = e + e(x-1) = ex$. Since $f''(x) = e^x > 0$, the exponential is convex and its curve lies above this tangent: $e^x \geq ex$ on $\mathbb{R}$.

E4. Set $g(x) = x^3 + 2x - 5$. This polynomial is continuous and $g'(x) = 3x^2 + 2 > 0$ on $\mathbb{R}$. Moreover $g(1) = -2 < 0$ and $g(2) = 7 > 0$: the IVT gives a zero in $]1, 2[$, and strict increase on $\mathbb{R}$ ensures that it is unique.

E5. After n steps, the length is $\frac{2}{2^n}$. The required condition is equivalent to $2^n \geq 2000$. Since $2^{10} = 1024 < 2000 \leq 2048 = 2^{11}$, eleven steps suffice and ten do not.

Solutions to Homework 4

1. $f(0) = 1$ and $\lim_{x\to 0^+} f(x) = 1$. At $+\infty$, $f(x) = xe^{-x} + e^{-x} \to 0$ by comparison of growth rates. By the product rule, $f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$. This derivative is strictly negative for $x > 0$: f strictly decreases from 1 towards 0 on $[0, +\infty[$.

2. $f''(x) = -e^{-x} + xe^{-x} = (x-1)e^{-x}$. So f is concave on $[0,1]$, convex on $[1, +\infty[$. The change of convexity at 1 gives the point of inflection $(1, \frac{2}{e})$.

3. f is continuous and strictly decreasing on $[0, +\infty[$; $f(0) = 1 > \frac{1}{2}$ and its limit 0 is less than $\frac{1}{2}$. The corollary of the IVT gives a unique solution α. Moreover, $f(1) = \frac{2}{e} \simeq 0.7358 > \frac{1}{2}$ and $f(2) = \frac{3}{e^2} \simeq 0.4060 < \frac{1}{2}$, so $1 < \alpha < 2$.

4.
```python
from math import exp

def racine_dm4():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+1)*exp(-m) > 0.5:
            a = m
        else:
            b = m
    return a, b
```

   The decrease preserves $f(a) \geq \frac{1}{2} \geq f(b)$; α stays in $[a,b]$. After eleven steps, we obtain $[1.677734375, 1.6787109375]$, of length $\frac{1}{1024} < 10^{-3}$. The value $\alpha \simeq 1.67835$ is consistent with this bracketing.

5. The tangent at 0 is $y = f(0) + f'(0)x = 1$. The variations give $f(x) \leq 1$, so the whole curve lies below this tangent. This does not characterise concavity: the curve would have to lie below all the tangents, whereas f becomes convex after 1.

Solutions to the textbook or past-paper work

For $f(x) = 5x^2e^x$, the product rule gives

$$f'(x) = 10xe^x + 5x^2e^x = 5x(x+2)e^x.$$

Since $e^x > 0$, f' is positive on $]-\infty;-2[$ and $]0;+\infty[$, negative on $]-2;0[$, zero at -2 and 0. The function therefore increases up to $f(-2) = 20e^{-2}$, decreases down to $f(0) = 0$, then increases. Its limits are 0 at $-\infty$ (comparison of growth rates) and $+\infty$ at $+\infty$. Differentiating again gives

$$f''(x) = 5(x^2+4x+2)e^x = 5\big((x+2)^2-2\big)e^x.$$

Set $a = -2-\sqrt{2}$ and $b = -2+\sqrt{2}$. The polynomial factor is positive outside $[a;b]$, negative in $]a;b[$. Thus f is convex on $]-\infty;a]$ and $[b;+\infty[$, concave on $[a;b]$. The sign changes at the two roots: the points of inflection are $(a;5a^2e^a)$ and $(b;5b^2e^b)$.

The table of variations is summarised as follows:

x	$-\infty$		-2		0		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	0	$\nearrow$	$20e^{-2}$	$\searrow$	0	$\nearrow$	$+\infty$

For question 5, the window below makes the two changes of convexity and the two extrema visible. The marked points correspond to the points of inflection calculated; the plot serves to check the consistency of the study.

